

# Andrew Alagna

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## Software Engineer | Frontend & Full-Stack Development | Accessibility & Performance

Results-driven Software Engineer with 3+ years of experience building high-performance, accessible (WCAG AAA) web applications at American Express. Skilled in React.js, Node.js, and modern JavaScript/TypeScript, with proven success driving measurable improvements in user experience, performance optimization, and secure account management for millions of global users. Strong collaborator and mentor with expertise in A/B testing, CI/CD pipelines, and scalable architecture, seeking to deliver impactful solutions in fast-paced NYC tech environments.

## PROFESSIONAL EXPERIENCE

### American Express

#### Software Engineer

08/2022 - Present

- Engineered secure user account management system using React.js, Node.js, and Kotlin, enabling millions of users to update sensitive information (name, address, contact details, income) while maintaining data integrity and regulatory compliance
- Spearheaded WCAG AAA accessibility compliance implementation across 10+ international markets using React.js, HTML5, CSS3, and JavaScript, improving accessibility audit scores from 72% to 99%, mitigating legal risk and expanding market reach for tens of millions of global users
- Enhanced web application performance by 10% through React.js optimization and BFF API architecture optimization, reducing average page load times across core products for consumers, small businesses, and corporate customers, resulting in improved customer satisfaction
- Drove user experience improvements by executing 15+ A/B experiments using JavaScript and analytics APIs, delivering measurable user experience improvements with 12% increase in completion rates and 15% improvement in customer satisfaction scores
- Built automated CI/CD pipelines using GitHub Actions, reducing deployment time by 15% and implementing feature flagging system with real-time toggles
- Collaborated in an Agile environment to deliver enterprise-scale applications. Drove quality by implementing a comprehensive TDD strategy with Jest/React Testing Library, achieving 100% code coverage and reducing production bugs

### CodePath

#### Intro to Software Engineering - Teaching Assistant

05/2022 - 08/2022

- Led instruction for over 180 students in Data Structures & Algorithms using Python, providing clear explanations and fostering understanding.
- Developed engaging problem-solving exercises, contributing to a 95% student satisfaction survey and demonstrating strong communication skills.
- Mentored students on technical concepts and career readiness, achieving measurable improvements in their performance and confidence.

### New York City Department of Transportation

#### Geographic Information Systems

04/2022 - 08/2022

- Engineered a Python application to optimize the processing and visualization of NYC's census data, improving data accuracy and accessibility for analysis.
- Streamlined data organization and labeling processes, reducing processing time from weeks to seconds for enhanced city-wide mapping.

## EDUCATION

### Bachelor of Arts in Computer Science

Hunter College • New York, NY • GPA: 3.62, Cum Laude

## VOLUNTEER LEADERSHIP

### CodePath

06/2021 - Present

#### Teaching & Mentoring

- Mentored and instructed over 200+ students across multiple cohorts in Data Structures & Algorithms, cybersecurity fundamentals, and career readiness, fostering an inclusive learning environment.
- Developed and presented engaging technical curriculum that contributed to 90%+ student satisfaction in end-of-course surveys, demonstrating strong communication and teaching skills.
- Guided students weekly through hands-on debugging, problem-solving, and technical interview preparation, contributing to their measurable improvement and securing free professional certifications for 100+ students.

## SKILLS

**Languages & Databases:** JavaScript (ES6+), PostgreSQL, Python, SQL, SQLite, TypeScript

**Frontend:** Component Libraries, CSS3, HTML5, React.js, Redux, Tailwind CSS

**Backend & APIs:** BFF Architecture, GraphQL, Kotlin, Node.js, REST

**DevOps & Tools:** AWS, Docker, Git, GitHub Actions (CI/CD), Jenkins

**Testing & Quality:** Jest, React Testing Library, TDD

**Core Expertise:** Accessibility (WCAG AAA), Agile/Scrum, Data Structures & Algorithms, Performance Optimization, System Design